KevaBook – System Workflow Specification

Purpose

This document describes how the KevaBook microservices collaborate to fulfill core business operations. It defines:

- Entry points (API endpoints)
- Participating microservices
- Database interactions
- Synchronous communication
- Asynchronous event processing
- Response generation

The workflows are based on the architecture and requirements specification.

Workflow 1: Business Owner Registration

Goal

Allow a business owner to create an account and to gain access to the platform.

Endpoint

POST /api/v1/users/register

Participating Services

- API Gateway
- User Service
- User Database

Workflow

Step 1 – Request Submission

The business owner submits registration information. For example:


```
{
  "firstName": "Nana",
  "lastName": "Appiah",
  "email": "nana@example.com",
  "password": "********"
}
```


Step 2 – API Gateway

The API Gateway:

- Receives the request
- Performs request logging
- Applies rate limiting
- Routes the request to User Service

Client
↓
API Gateway
↓
User Service

Step 3 – User Service Validation

User Service:

- Validates required fields
- Validates email format
- Checks password requirements

Step 4 – Duplicate User Check

User Service queries User Database.

SELECT user WHERE email = ?

If email already exists:

409 Conflict

Step 5 – Account Creation

User Service:

- Hashes password
- Creates Business Owner account
- Persists user record

Step 6 – Database Persistence

User information is stored in User Database.

Step 7 – Response

201 Created

```
{
  "ownerId": "123",
  "email": "nana@example.com"
}
```

Responsibilities

User Service

Responsible for:

- Registration
- Credential management
- Authentication-related user data

Workflow 2: Business Profile Creation

Goal

Allow a registered business owner to create a business profile.

Endpoint

POST /api/v1/businesses

Participating Components

- API Gateway
- User Service
- Business Service
- Business Database

Workflow

Step 1 – Request Submission

Authenticated business owner submits business information.

Step 2 – API Gateway

Gateway validates JWT token and forwards request.

Step 3 – Business Service Authorization

Business Service extracts owner information from JWT.

Step 4 – Business Profile Validation

Business Service validates:

- Business name
- Contact details
- Required fields

Step 5 – Persist Business Profile

Business profile is stored in Business Database.

Step 6 – Response

201 Created


```
{
  "businessId": "456",
  "name": "Nana Consulting"
}
```


Responsibilities

Business Service

Responsible for:

- Business profile lifecycle
- Business settings
- Business ownership validation

Workflow 3: Service Creation

Goal

Allow business owners to create bookable services.

Endpoint

POST /api/v1/services

Participating Components

- API Gateway
- Service Catalog Service
- Business Service
- Service Database

Workflow

Step 1 – Request Submission

Business owner submits service details.


```
{
  "businessId": "456",
  "name": "Career Coaching",
  "duration": 60,
  "price": 50
}
```


Step 2 – Service Validation

Service Catalog Service validates:

- Name
- Duration
- Price

Step 3 – Business Verification

Service Catalog Service verifies that:

- Business exists
- Authenticated owner owns the business

Request sent to:

GET /internal/businesses/{id}

Step 4 – Service Persistence

Service is saved to the Service Database.

Step 5 – Response

201 Created

```
{
  "serviceId": "789"
}
```

Responsibilities

Service Catalog Service

Responsible for:

- Service lifecycle management
- Service catalog maintenance
- Service search operations

Workflow 4: Availability Retrieval

Goal

Allow clients to view available appointment slots before booking.

Endpoint

GET /api/v1/businesses/{businessId}/availability

Participating Components

- API Gateway
- Booking Service
- Service Catalog Service
- Booking Database

Workflow

Step 1 – Request Submission

Client requests available slots.

Step 2 – Booking Service Retrieves Availability Rules

Booking Service loads:

- Working hours
- Availability schedules

Step 3 – Booking Service Retrieves Existing Bookings

Booking Service queries Booking Database.

Purpose:

- Identify occupied slots
- Eliminate unavailable times

Step 4 – Slot Generation

Booking Service dynamically generates available slots.

Example:

09:00
10:00
11:00
14:00
15:00

Step 5 – Response


```
[
  {
    "date": "2026-06-10",
    "startTime": "09:00"
  }
]
```


Responsibilities

Booking Service

Responsible for:

- Availability management
- Slot generation
- Booking conflict detection

Workflow 5: Booking Creation

Goal

Allow a client to create an appointment booking.

Endpoint

POST /api/v1/bookings

Participating Components

- API Gateway
- Booking Service
- Service Catalog Service
- Booking Database
- RabbitMQ
- Notification Service
- Notification Database

Workflow

Step 1 – Request Submission

Client submits booking request.


```
{
  "serviceId": "789",
  "bookingDate": "2026-06-10",
  "startTime": "10:00",
  "clientName": "John Doe",
  "clientEmail": "john@example.com"
}
```

Step 2 – Validation

Booking Service validates:

- Service exists
- Booking date is valid
- Client information is complete

Step 3 – Service Lookup

Booking Service requests service details.

GET /internal/services/{serviceId}

Service Catalog Service returns:

- Duration
- Business ID
- Active status

Step 4 – Availability Check

Booking Service verifies:

- Slot is available
- Slot is not already booked

Step 5 – Booking Creation

Booking Service:

- Creates Booking ID
- Generates Booking Reference
- Persists booking

Step 6 – Event Publication

Booking Service publishes: BOOKING_CREATED to RabbitMQ.

Step 7 – Notification Processing

Notification Service consumes event.

Notification Service:

- Creates confirmation email
- Sends email to client
- Stores notification history

Step 8 – Response

201 Created

```
{
  "bookingId": "ABC123",
  "bookingReference": "KVB-10293",
  "status": "CONFIRMED"
}
```

Responsibilities

Booking Service

Responsible for:

- Booking lifecycle
- Availability validation
- Booking integrity
- Event publication

Notification Service

Responsible for:

- Email delivery
- SMS delivery (future)
- Notification history

Workflow 6: Booking Cancellation

Goal

Allow clients or business owners to cancel appointments.

Endpoint

DELETE /api/v1/bookings/{bookingReference}

Participating Components

- API Gateway
- Booking Service
- RabbitMQ
- Notification Service

Workflow

Step 1 – Cancellation Request

Client or business owner requests cancellation.

Step 2 – Booking Validation

Booking Service:

- Finds booking
- Verifies status
- Verifies cancellation eligibility

Step 3 – Status Update

Booking status updated to: CANCELLED

Step 4 – Event Publication

Booking Service publishes: BOOKING_CANCELLED

Step 5 – Notification Processing

Notification Service:

- Sends cancellation email
- Records notification history

Step 6 – Response

200 OK

```
{
  "status": "CANCELLED"
}
```

Workflow 7: Notification Delivery

Goal

Process domain events and deliver notifications asynchronously.

Triggering Events

- BOOKING_CREATED
- BOOKING_CANCELLED
- BOOKING_REMINDER

Participating Components

- RabbitMQ
- Notification Service
- SMTP Provider
- Notification Database

Workflow

Step 1 – Event Consumption

Notification Service consumes domain event.

Step 2 – Template Resolution

Notification Service selects the appropriate template.

Examples:

- Booking Confirmation
- Booking Cancellation
- Booking Reminder

Step 3 – Message Generation

Notification Service generates notification content.

Step 4 – Delivery

Notification Service sends email through SMTP provider.

Future:

SMS Provider

Step 5 – Persistence

Notification status is stored.

PENDING
SENT
FAILED

Step 6 – Monitoring

Notification metrics and delivery status become available for observability dashboards.

Service Responsibility Summary

Services	Key Primary Responsibilities
User survive	Business owner registration, for authentication, profile management
Business service	Business profile, ownership, business settings
Service catalog service	Service catalog management, service lookup,service search

Booking Service	Availability, slot generation, booking lifecycle, booking integrity
Notification Service	Email delivery, reminders, notification history
API Gateway	Routing, security, rate limiting, request logging
RabbitMq	Event distribution and asynchronous communication